LMS Seminar

Dynamic rupture behavior and friction evolution revealed by laboratory experiments using ultra high-speed digital image correlation

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Venue: Amphi 104 (Pole Meca)

Abstract

Shear cracks along interfaces are relevant to a broad class of engineering and geophysics applications, ranging from the failure of composite materials and bonded joints to earthquakes and landslides. Characterizing the rheology of interfaces and faults is of paramount importance to improve our understanding of frictional ruptures, as friction controls key processes of rupture nucleation, propagation, and arrest and influences various rupture outcomes, for example how damaging earthquakes can be. In this presentation, I will discuss recent advances in the characterization of rupture behavior and friction evolution using our newly developed imaging technique, based on digital image correlation (DIC) coupled with ultrahigh speed photography. One of the highlights of this new approach is its ability to experimentally capture the full-field evolution of particle velocities and strains of spontaneously propagating dynamic ruptures achieving a level of detail that was previously only possible through numerical simulations. Dynamic imaging of stresses enables us to decode the nature of friction by tracking its evolution and studying its dependence on slip, slip velocity and their history. The measured friction behavior allows us to challenge existing friction laws and formulate new ones. This approach gives a new perspective on the study of friction and provides important insights into earthquake and rupture physics.

About the speaker

Vito Rubino is Associate Professor at École Centrale de Nantes, France, since September 2022. His research interests focus on experimental and computational mechanics to study fracture and friction phenomena. Dr Rubino received his undergraduate degree from Politecnico di Torino, Italy, while holding a research assistant position at Imperial College London on his final year. He obtained his Ph.D. from the University of Cambridge in 2008. After a brief experience in industry on R& D with Airbus UK, in 2011 he started his postdoctoral studies at the California Institute of Technology (Caltech), in the Department of Aerospace (GALCIT). He then became Research Scientist at Caltech in 2015. He recently earned a NeXT Talent award at Centrale Nantes.

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